

Machine-Learning Prediction of Aqueous Molecular Solubility Using Molecular Descriptors

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Introduction

Aqueous solubility is an important physicochemical property that influences the behavior of chemical compounds in pharmaceutical development, environmental transport, separation processes, and chemical product design. Experimental determination of solubility can be time-consuming and resource-intensive; therefore, computational models capable of estimating solubility directly from molecular structure are valuable screening tools. The objective of this project was to develop and evaluate machine-learning models for predicting the experimentally measured logarithmic aqueous solubility of organic compounds from their SMILES representations. The workflow included molecular structure processing, descriptor calculation, data preprocessing, exploratory analysis, feature selection, model development, hyperparameter optimization, and prediction of solubility for previously unseen molecules.

Dataset

The project used the Delaney ESOL dataset, consisting of 1,144 organic compounds. Each observation included a compound identifier, a SMILES string, and an experimentally measured aqueous solubility value expressed as $\log(\text{solubility:mol/L})$. The measured logarithmic solubility was treated as the continuous target variable. The SMILES strings were used to reconstruct molecular structures and generate numerical molecular descriptors for use as model inputs.

Molecular Representation and Descriptor Calculation

SMILES strings were converted into RDKit molecular objects. Each input structure was validated to identify invalid SMILES representations. Explicit hydrogen atoms were subsequently added, and three-dimensional molecular conformations were generated. Conformer-generation failures were also checked to prevent invalid molecular structures from being passed to the descriptor-calculation stage.

Molecular descriptors were calculated using Mordred with both two-dimensional and three-dimensional descriptor calculation enabled. These descriptors numerically represented molecular characteristics such as size, atomic composition, surface-area distribution, charge distribution, connectivity, and structural complexity.

Data Preprocessing

The descriptor dataset contained numerical, Boolean, and unsupported object-type variables. Object-type descriptors were excluded because they could not be directly used by the regression models. Boolean descriptors retained in the analysis were converted to integer representations. Descriptors with constant values were removed because they provided no discriminatory information. To reduce redundancy and multicollinearity, highly correlated variables were filtered using an absolute correlation threshold of 0.80. Constant and correlated descriptors were identified using the training portion of the dataset in order to minimize information leakage from the test set. The dataset was divided into training and testing subsets using an 80:20 ratio and a fixed `random_state` of 45.

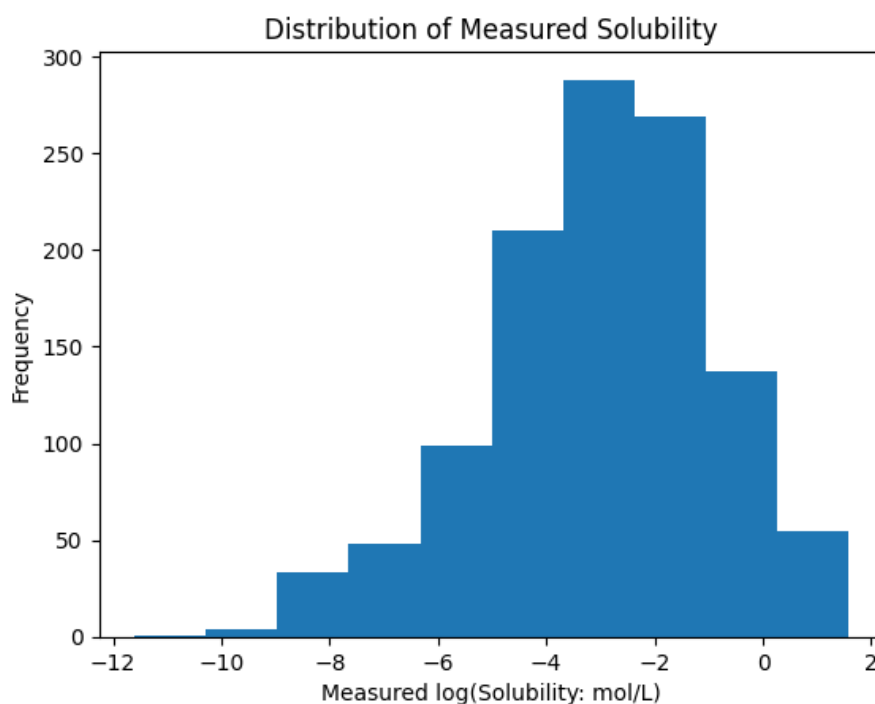


Figure 1: Distribution of experimentally measured aqueous solubility values in the Delaney ESOL dataset.

The distribution shows the range and frequency of solubility values represented in the dataset and provides context for the regression task. Standardization was applied by fitting a `StandardScaler` exclusively to the training data and subsequently transforming both the training and testing sets with the fitted scaler.

Exploratory Data Analysis and Feature Selection

Exploratory analysis included descriptive statistics, visualization of the target distribution, correlation analysis, scatter plots, principal component analysis, and t-distributed stochastic neighbor embedding. These analyses were used to examine the distribution of measured solubility, relationships between descriptors, potential molecular groupings, and the dimensional structure of the feature space. Several feature-importance methods were also applied, including Random Forest impurity-based importance, permutation importance, and SHAP analysis. These methods were used to evaluate the contribution of individual descriptors to model predictions.

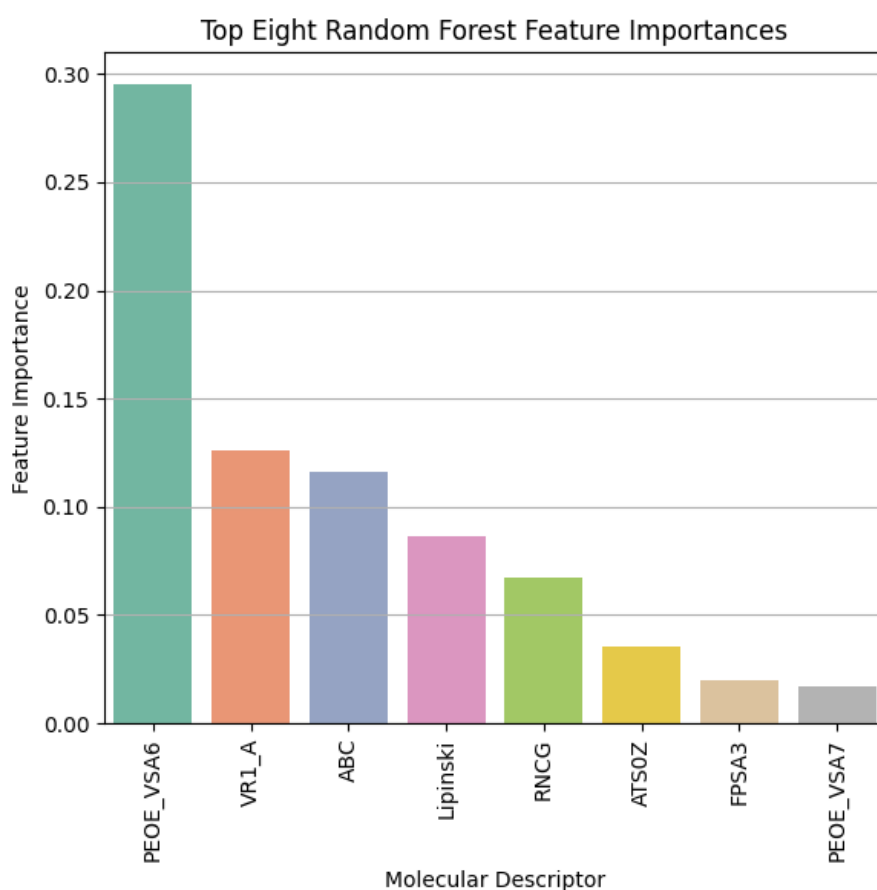


Figure 2: Relative importance of the molecular descriptors according to the Random Forest model.

The final reduced dataset contained eight selected molecular descriptors:

- **PEOE_VSA6:** A two-dimensional Molecular Operating Environment (MOE) charge-surface-area descriptor. It represents the sum of the van der Waals surface areas of

atoms whose partial charges fall within the range $-0.10 \leq q < -0.05$. It therefore captures the contribution of moderately negatively charged atomic regions to the molecular surface.

- **VR1_A:** A two-dimensional topological descriptor derived from the molecular adjacency matrix. It is based on the eigenvalues and eigenvectors of the matrix and provides information about molecular connectivity and overall graph structure.
- **ABC:** The atom–bond connectivity index, a two-dimensional topological descriptor calculated from the degrees of bonded atoms. It reflects molecular branching and connectivity and generally increases with molecular size and structural complexity.
- **Lipinski:** A Boolean descriptor indicating whether a molecule satisfies Lipinski’s Rule of Five. In Mordred, the descriptor is true when the molecule has a molecular weight no greater than 500, a calculated $\log P$ no greater than 5, no more than five hydrogen-bond donors, and no more than ten hydrogen-bond acceptors.
- **RNCG:** The relative negative charge descriptor. It expresses the magnitude of the most negatively charged atom relative to the total negative partial charge of the molecule, thereby describing how concentrated or distributed the negative charge is across the molecular structure.
- **ATS0Z:** A two-dimensional Moreau–Broto autocorrelation descriptor at lag zero, weighted by atomic number. Because the lag is zero, it primarily summarizes the atomic-number composition of the molecule rather than relationships between atoms separated by several bonds.
- **FPSA3:** The version-3 fractional charged partial positive surface-area descriptor. It relates the positively charged partial surface area to the total molecular surface area and therefore represents the fraction of the molecular surface associated with positive partial charge.
- **PEOE_VSA7:** A two-dimensional MOE charge–surface-area descriptor representing the sum of the van der Waals surface areas of atoms whose partial charges fall within the range $-0.05 \leq q < 0.00$. It describes molecular surface regions associated with weakly negative or nearly neutral atoms.

These descriptors represented different aspects of molecular surface area, molecular connectivity, charge distribution, graph structure, and drug-like molecular properties.

Model Development

Multiple regression approaches were evaluated during model development. Linear Regression was first used as a baseline model. It achieved an (R^2) value of 0.954 on the training set and 0.754 on the test set. The difference between these values indicated that the linear model did not generalize as effectively to unseen compounds and suggested that the relationship between the molecular descriptors and solubility was not fully linear.

A Random Forest Regressor trained on the larger descriptor set produced an (R^2) of 0.982 on the training set and 0.863 on the test set. The corresponding test mean absolute error was 0.553 log mol/L, while the root mean squared error was 0.770 log mol/L. Five-fold cross-validation produced a mean (R^2) of approximately 0.869, indicating relatively stable predictive performance across the training folds.

Principal component regression was also examined. The first eight principal components explained approximately 42.18% of the total descriptor variance. Because a substantial proportion of the molecular information remained outside these components, PCA was retained primarily as an exploratory dimensionality-reduction method rather than the final predictive approach.

Random Forest and Extra Trees regressors were selected for further evaluation because ensemble tree methods are capable of modelling nonlinear relationships and interactions between molecular descriptors. They are also relatively robust to complex feature distributions and provide interpretable measures of feature importance.

Hyperparameter Optimization and Final Models

The models were evaluated using the reduced eight-descriptor dataset. Before hyperparameter optimization, the Random Forest model achieved a test (R^2) of approximately 0.832, while the Extra Trees model achieved a test (R^2) of approximately 0.836.

Grid search was used to optimize the Random Forest and Extra Trees hyperparameters. The optimized Random Forest grid-search model used 700 trees, no bootstrap

sampling, unrestricted tree depth, square-root feature selection, a minimum leaf size of 2, and a minimum split size of 5. This model achieved a test (R^2) of approximately 0.825.

The optimized Extra Trees model used 400 trees, no bootstrap sampling, a maximum depth of 30, logarithmic feature selection, a minimum leaf size of 2, and a minimum split size of 5. Its test (R^2) was approximately 0.807.

Randomized search was additionally applied to the Random Forest model. The final synchronized evaluation of the reduced-feature models produced the following results:

Table 1

Model	Training (R^2)	Testing (R^2)	Test MAE	Test RMSE
Random Forest	0.950	0.809	0.636	0.909
Extra Trees	0.946	0.807	0.645	0.913

The final Random Forest model demonstrated slightly better performance than the Extra Trees model. However, the predictive results of the two ensemble approaches were highly similar.

Prediction of New Compounds

Prediction functions were developed to estimate solubility from either a single SMILES string or a list of SMILES strings. Each function validated the molecular input, generated the required descriptors, selected the features in the same order used during model training, applied the fitted scaler, and generated predictions using both final ensemble models. For the example SMILES string OCC1=C(O)C=C(O)C=C1, the Random Forest model predicted a solubility of approximately -1.415 log mol/L, while the Extra Trees model predicted approximately -1.264 log mol/L.

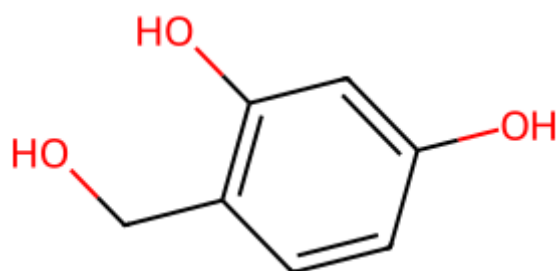


Figure 3: Two-dimensional molecular structure generated from the example SMILES string used to demonstrate the prediction function.

Discussion

The results demonstrate that molecular descriptors derived from SMILES strings can be used to predict aqueous solubility with reasonably strong performance. Ensemble tree models substantially outperformed the Linear Regression baseline on the held-out test set, indicating that nonlinear relationships and interactions between molecular properties were important for predicting solubility.

The Random Forest model trained on the larger descriptor set achieved the highest reported test performance, with an (R^2) of 0.863. Reducing the input space to eight descriptors resulted in a modest decline in predictive performance. Nevertheless, the reduced models retained useful predictive ability while improving simplicity, interpretability, and computational efficiency.

The difference between training and test performance indicated that some degree of overfitting remained. Furthermore, the final performance estimates were based primarily on one random train-test split. The study did not include an independent external validation dataset or an explicit applicability-domain analysis. Consequently, predictions for molecules that differ substantially from the chemical space represented in the ESOL dataset should be interpreted cautiously.

Conclusion

This project established a complete machine-learning workflow for predicting aqueous molecular solubility from SMILES strings. RDKit and Mordred were used to convert molecular structures into numerical descriptors, after which preprocessing and feature-selection methods were applied to produce a reliable modelling dataset.

Among the evaluated approaches, ensemble tree methods provided the strongest predictive performance. The Random Forest model trained on the broader descriptor set achieved the highest test (R^2), while reduced-feature Random Forest and Extra Trees models provided simpler alternatives with comparable performance.

Future work should investigate scaffold-based or repeated cross-validation, independent external validation, molecular fingerprint representations, applicability-domain assessment, prediction uncertainty, and graph-based deep-learning methods. These extensions would provide a more rigorous evaluation of model generalizability and could improve predictive performance across a broader chemical space.